

Ferroelectric relaxation dynamics of twisted bilayer hexagonal boron nitride

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2D van der Waals heterostructures is a fascinating group of materials, due to the control and tunability of properties. This can be done in multiple ways, but a relatively new way of tuning these materials is twisting of a layer. Examples of the effect of twisting include superconductivity in graphene and ferroelectricity in hexagonal boron nitride [1, 2]. There is a relatively new theory of sliding ferroelectricity, in which sliding in xy-plane leads to a polarization in z-direction [3]. Effectively this perturbation in xy-plane can be seen to lead to a distortion of the pz-orbital, i.e., the charge distribution in that direction. Rotation (or twisting) and sliding are similar enough that this could be a part of the explanation for the emerging polarization. Studying the lattice dynamics of this system is crucial to understanding the material properties. Here we study the relaxation displacements and phonons (lattice vibrations) in order to gain more information on material, keeping the transition from insulating to ferroelectric phase, that results from twisting, in mind.

[1] Cao, Y., Fatemi, V., Fang, S. et al, *Nature* **556**, <https://doi.org/10.1038/nature26160> (2018).

[2] Kenji Yasuda et al., *Science* **372**, 10.1126/science.abd3230 (2021).

[3] Junyi Ji, Guoliang Yu, Changsong Xu, and H. J. Xiang, *Phys. Rev. Lett.* **130**, <https://doi.org/10.1103/PhysRevLett.130.146801> (2023).